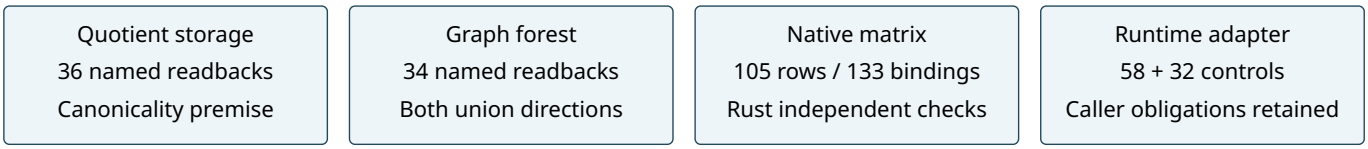
Storage, forest and native verification interfaces

This packet completes additional source-model connections for quotient storage and graph forest preservation, and adds native implementations for matrix verification and runtime diff qualification predicates. Producer sources are published before independent reviews. The frozen 1,260-target scientific scope remains unchanged.



Quotient storage. The executable model initializes an Array UInt64, performs indexed coefficient writes, follows the existing remainder loop and trims the final quotient. Whole-loop refinement connects the stored words to the polynomial quotient and exact-division endpoint. Producer checks cover 15 main declarations, 13 producer controls and 8 independent controls, with 36 named axiom readbacks. Allocation arithmetic excludes unsigned length-plus-one overflow under explicit nonempty-divisor and representable-input assumptions; physical allocation, byte-capacity and compiled usize remain separate.

Graph forest. Parent-path reachability defines the component whose rank changes during attachment. The proof preserves forest structure, root paths and global vertex/anchor equations through two finds and either union direction, then derives a finite next find. Producer checks cover 10 main declarations, 9 producer controls and 15 independent controls, with 34 named readbacks. Native size metadata, allocation, word/array bounds and complete same-root edge composition remain explicit interfaces.

Native matrix verification. The prior Python independent verifier violated the repository language rule and is preserved as historical evidence. Its replacement is Rust, with widened dot products and extended-Euclid normalization. It reproduces 105 ordered saved rows and enforces the fixed 133-file source binding before and after computation. Ten native controls, eight independent Rust boundary tests and two actual CLI corruption cases pass; a substituted walk source or changed binding manifest is rejected. The original native producer remains unchanged.

Native runtime adapter. Rust qualify checks the identity, terminal state, mount, security and resource prerequisites before exact diff parsing. Fifty-eight producer controls and 32 independent controls pass their expected outcomes. Duplicate identical tmpfs options and matching absolute paths containing parent segments are accepted; caller filesystem canonicalization remains required. The initial Python draft was never executed and is retained as a rejected direction. The actual full native supervisor/callsite remains unported; the binding template is unadmitted and supplies no runtime or campaign authorization.

Scope and custody. Current main 3e424214a63121afc3575750dfeb1f76b5a04064 has byte-identical three_leads.rs. The shared-record merge conflict and journal reservation remain separate. No scientific job, Docker qualification, point enumeration, timing or new cost ratio was performed. Sources, native failures, axiom readbacks and reviewer findings are retained with exact hashes. Native compiled build trees and copied CLI negative-input trees are omitted from publication; their hashes and check.sh reconstruction remain in the producer manifest and fixed binding.

Independent review boundaries. Quotient canonicality is a theorem premise, not runtime rejection of noncanonical words. The graph review includes an actual same-root second find, while the complete native same-root mutation branch remains separate. The native matrix reviewer discloses authorship of the earlier producer; the adapter reviewer discloses prior supervisor familiarity. Reviews followed producer publication at 09e64fca55.

Native verification correction. The formal reviewers initially used Python acceptance and replay wrappers, contrary to the repository rule. Their records are preserved. Additive native verifiers and Rust or thin-shell runners freshly replayed all 8 quotient modules and 10 graph modules, retaining 36 and 34 named readbacks respectively. Deliberately false theorems failed with exit 1; altered readback names and malformed records were rejected. Final acceptance uses these native paths.